

Multiple Smooth Terms

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Table of contents

Overview	1
Setup	2
Loading Gu–Wahba data	2
Fitting the model	3
Model overview	3
Plotting each smooth	5
Concurvity	6
Comparing models with different k	8
Summary	8

Overview

One of the key strengths of GAMs is the ability to include **multiple smooth terms** additively. The model:

$$y_i = \beta_0 + f_0(x_{0i}) + f_1(x_{1i}) + f_2(x_{2i}) + f_3(x_{3i}) + \varepsilon_i$$

estimates each smooth function f_j simultaneously, with automatic smoothness selection for each term.

This vignette uses the classic **Gu–Wahba** simulation (equivalent to `gamSim(eg = 1)` in `mgcv`) with four smooth functions of varying complexity.

Setup

```
library(mgcv)
```

Loading required package: nlme

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.

```
library(gratia)
library(ggplot2)
```

Loading Gu–Wahba data

The four test functions are:

- $f_0(x) = 2 \sin(\pi x)$ — a smooth sinusoid
- $f_1(x) = e^{2x}$ — an exponential
- $f_2(x) = 0.2x^{11}(10(1-x))^6 + 10(10x)^3(1-x)^{10}$ — a bumpy function
- $f_3(x) = 0$ — a null smooth (no effect)

```
df <- read.csv("../data.csv")
n <- nrow(df)
x0 <- df$x0; x1 <- df$x1; x2 <- df$x2; x3 <- df$x3
y <- df$y

f0 <- function(x) 2 * sin(pi * x)
f1 <- function(x) exp(2 * x)
f2 <- function(x) 0.2 * x^11 * (10 * (1 - x))^6 + 10 * (10 * x)^3 * (1 - x)^10
f3 <- function(x) rep(0, length(x))

# Center the true functions for identifiability
f0_vals <- f0(x0); f0_vals <- f0_vals - mean(f0_vals)
f1_vals <- f1(x1); f1_vals <- f1_vals - mean(f1_vals)
f2_vals <- f2(x2); f2_vals <- f2_vals - mean(f2_vals)
f3_vals <- f3(x3)

head(df)
```

	y	x0	x1	x2	x3
1	-4.4742515	0.9148060	0.02270001	0.9090475	0.4018804
2	-2.7697245	0.9370754	0.51323953	0.8999248	0.4322142
3	6.4688919	0.2861395	0.63072615	0.1923493	0.6636044
4	-1.7536068	0.8304476	0.41877162	0.5322903	0.1823693
5	0.1671775	0.6417455	0.87926595	0.5221247	0.8383388
6	2.3332107	0.5190959	0.10798707	0.1603357	0.9173730

Fitting the model

```
m <- gam(y ~ s(x0) + s(x1) + s(x2) + s(x3), data = df, method = "REML")
summary(m)
```

Family: gaussian

Link function: identity

Formula:

$y \sim s(x_0) + s(x_1) + s(x_2) + s(x_3)$

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.02824	0.10505	0.269	0.788

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(x0)	3.425	4.244	8.828	8.78e-07 ***
s(x1)	3.221	4.003	67.501	< 2e-16 ***
s(x2)	7.905	8.685	67.766	< 2e-16 ***
s(x3)	1.885	2.359	2.642	0.0636 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.685 Deviance explained = 69.8%

-REML = 886.93 Scale est. = 4.4144 n = 400

Model overview

The `gratia::overview()` function summarizes all smooth terms:

```
overview(m)
```

Generalized Additive Model with 4 terms

	term	type	k	edf	statistic	p.value
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<chr>
1	s(x0)	TPRS	9	3.42	8.83	< 0.001
2	s(x1)	TPRS	9	3.22	67.5	< 0.001
3	s(x2)	TPRS	9	7.90	67.8	< 0.001
4	s(x3)	TPRS	9	1.88	2.64	0.063622

Per-smooth EDF:

```
cat("Per-smooth EDF:\n")
```

Per-smooth EDF:

```
for (i in seq_along(m$smooth)) {  
  cat(sprintf("  %-12s EDF = %.2f\n", m$smooth[[i]]$label, pen.edf(m)[i]))  
}
```

s(x0)	EDF = 3.42
s(x1)	EDF = 3.22
s(x2)	EDF = 7.90
s(x3)	EDF = 1.88

```
cat(sprintf("\nTotal model EDF: %.2f\n", sum(m$edf)))
```

Total model EDF: 17.44

```
cat(sprintf("Deviance explained: %.1f%%\n", summary(m)$dev.expl * 100))
```

Deviance explained: 69.8%

Plotting each smooth

We plot each smooth estimate alongside the corresponding true function:

```
true_fns <- list(f0, f1, f2, f3)
x_vars <- c("x0", "x1", "x2", "x3")
x_data <- list(x0, x1, x2, x3)
f_centered <- list(f0_vals, f1_vals, f2_vals, f3_vals)

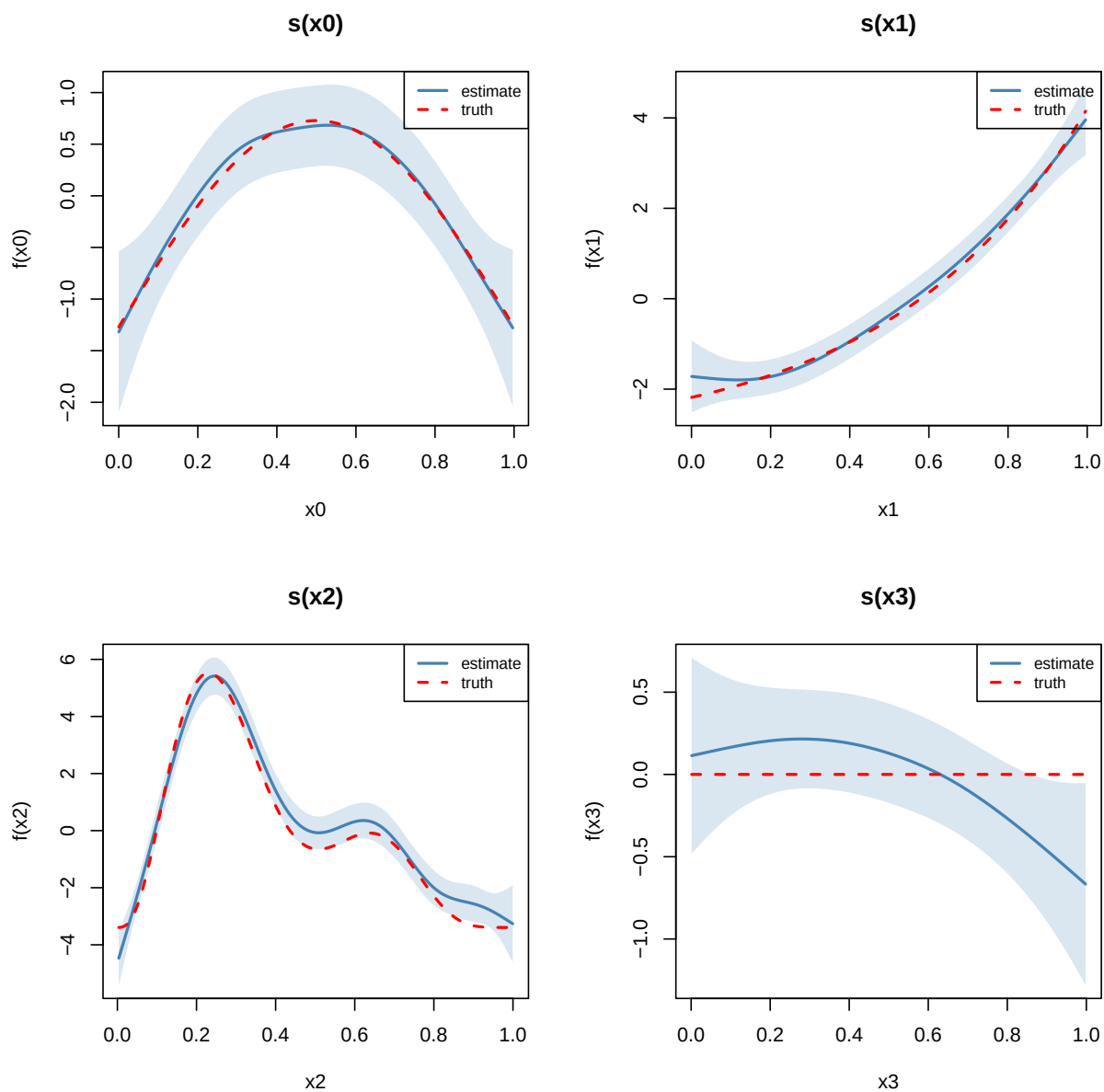
par(mfrow = c(2, 2))
for (i in 1:4) {
  se <- smooth_estimates(m, smooth = paste0("s(", x_vars[i], ")"), n = 200)
  x_grid <- se[[x_vars[i]]]

  # True function on grid, centered
  f_grid <- true_fns[[i]](x_grid)
  f_grid <- f_grid - mean(f_grid)

  plot(x_grid, se[[".estimate"]], type = "l", lwd = 2, col = "steelblue",
       ylim = range(c(se[[".estimate"]] - 2 * se[[".se"]], se[[".estimate"]] + 2 * se[[".se"]]),
       xlab = x_vars[i], ylab = paste0("f(", x_vars[i], ")"),
       main = paste0("s(", x_vars[i], ")"))
  polygon(c(x_grid, rev(x_grid)),
         c(se[[".estimate"]] - 2 * se[[".se"]], rev(se[[".estimate"]] + 2 * se[[".se"]])),
         col = adjustcolor("steelblue", alpha.f = 0.2), border = NA)
  lines(x_grid, f_grid, lty = 2, lwd = 2, col = "red")
  legend("topright", c("estimate", "truth"), lty = c(1, 2),
       col = c("steelblue", "red"), lwd = 2, cex = 0.8)
}
```

Warning: The `smooth` argument of `smooth\_estimates()` is deprecated as of gratia 0.8.9.9.

i Please use the `select` argument instead.



```
par(mfrow = c(1, 1))
```

Note that $f_3(x_3) = 0$ (the null smooth), and the model should estimate it as approximately flat with an EDF close to 0–1.

Concurvity

Concurvity is the smooth analogue of collinearity — it measures how well each smooth can be approximated by the other smooth terms. Values close to 1 indicate potential confounding.

```
conc <- concavity(m, full = TRUE)
print(round(conc, 3))
```

```
      para s(x0) s(x1) s(x2) s(x3)
worst      0 0.127 0.130 0.131 0.169
observed   0 0.042 0.053 0.052 0.065
estimate   0 0.082 0.060 0.081 0.052
```

Since our covariates are independent (uniform draws), concavity should be low.

Pairwise concavity:

```
conc_pw <- concavity(m, full = FALSE)
print(lapply(conc_pw, function(x) round(x, 3)))
```

```
$worst
      para s(x0) s(x1) s(x2) s(x3)
para      1 0.000 0.000 0.000 0.000
s(x0)     0 1.000 0.059 0.086 0.060
s(x1)     0 0.059 1.000 0.056 0.076
s(x2)     0 0.086 0.056 1.000 0.084
s(x3)     0 0.060 0.076 0.084 1.000
```

```
$observed
      para s(x0) s(x1) s(x2) s(x3)
para      1 0.000 0.000 0.000 0.000
s(x0)     0 1.000 0.023 0.021 0.010
s(x1)     0 0.014 1.000 0.014 0.027
s(x2)     0 0.014 0.012 1.000 0.027
s(x3)     0 0.014 0.015 0.018 1.000
```

```
$estimate
      para s(x0) s(x1) s(x2) s(x3)
para      1 0.000 0.000 0.000 0.000
s(x0)     0 1.000 0.029 0.019 0.006
s(x1)     0 0.033 1.000 0.023 0.017
s(x2)     0 0.040 0.011 1.000 0.027
s(x3)     0 0.011 0.015 0.039 1.000
```

Comparing models with different k

Does the basis dimension k matter? Let us refit with smaller and larger k:

```
for (k_val in c(5, 10, 20, 30)) {  
  mk <- gam(y ~ s(x0, k = k_val) + s(x1, k = k_val) +  
            s(x2, k = k_val) + s(x3, k = k_val),  
            data = df, method = "REML")  
  edfs <- round(pen.edf(mk), 2)  
  dev <- round(summary(mk)$dev.expl * 100, 1)  
  cat(sprintf("k=%d  EDF=[%s]  Dev.Expl=%.1f%%\n",  
              k_val, paste(edfs, collapse = ", "), dev))  
}
```

```
k=5  EDF=[2.98, 3.06, 3.97, 1.47]  Dev.Expl=66.0%  
k=10 EDF=[3.42, 3.22, 7.9, 1.88]  Dev.Expl=69.8%  
k=20 EDF=[3.45, 3.25, 9.99, 1.87]  Dev.Expl=70.2%  
k=30 EDF=[3.46, 3.25, 10.23, 1.87] Dev.Expl=70.2%
```

As long as k is large enough to capture the true complexity, results are similar — the penalty takes care of the rest.

Summary

In this vignette we:

1. Simulated Gu–Wahba data with four smooth functions of varying complexity
2. Fitted a GAM with four additive smooth terms
3. Examined per-smooth EDF via `overview()`
4. Plotted each smooth estimate against the true function
5. Assessed concurrency between smooth terms
6. Demonstrated robustness to the choice of k

The next vignette covers non-Gaussian response distributions.